

SN 2023vbg : A 09ip-like Type II_n Supernova with Precursor Activity and Early Bump

Sota Goto (Kagoshima University)

Supervisor : Takahiro Nagayama, Masayuki Yamanka

This talk is based on our recent paper:

Goto et al. (2025), *SN 2023vbg: A Type II_n Supernova Resembling SN 2009ip, with a Long-duration Precursor and Early-time Bump*



Outline of this talk

- Iriki Observatory & Our Group
 - introduction our group
- What is a 09ip-like supernova?
- SN 2023vbg
 - Discovery and basic properties
- Light curve evolution
- Blackbody fitting (SED evolution)
- Spectral evolution
- Discussion
 - mass-loss, comparison with 09ip-like SNe
 - Nature of SN 2023vbg

Outline of this talk

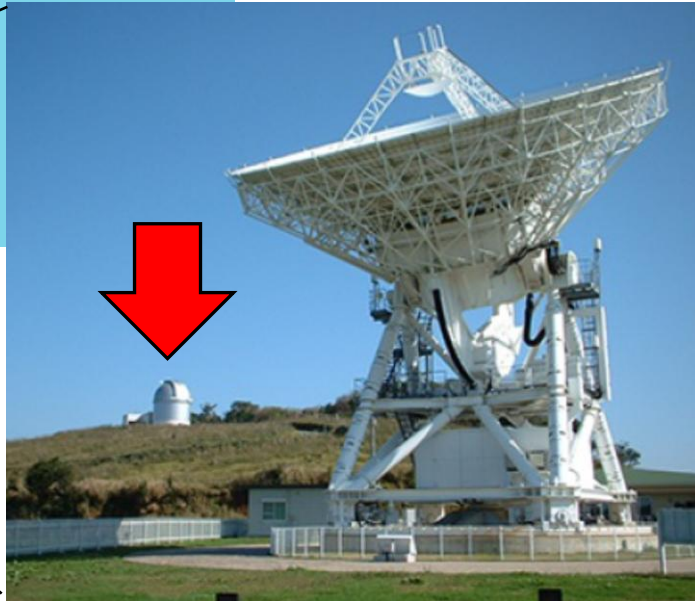
- Iriki Observatory & Our Group
 - introduction our group
- What is a 09ip-like supernova?
- SN 2023vbg
 - Discovery and basic properties
- Light curve evolution
- Blackbody fitting (SED evolution)
- Spectral evolution
- Discussion
 - mass-loss, comparison with 09ip-like SNe
 - Nature of SN 2023vbg

Iriki Observatory & Our Group



Kagoshima university 1m telescope

- 1.0m optical and NIR telescope
- Located at Iriki Observatory, Kagoshima
- **Flexible scheduling and rapid response**
- kSIRIUS + optcam (g,i-band)
⇒ **5-bands simultaneous imaging**



Main instrument : kSIRIUS

- Near-infrared imager
- **Simultaneous JHKs bands imaging**
- Typical limiting magnitudes:
J~19, H~18, Ks~16 mag (2h)

Iriki Observatory & Our Group

Nagayama group (Kagoshima University)

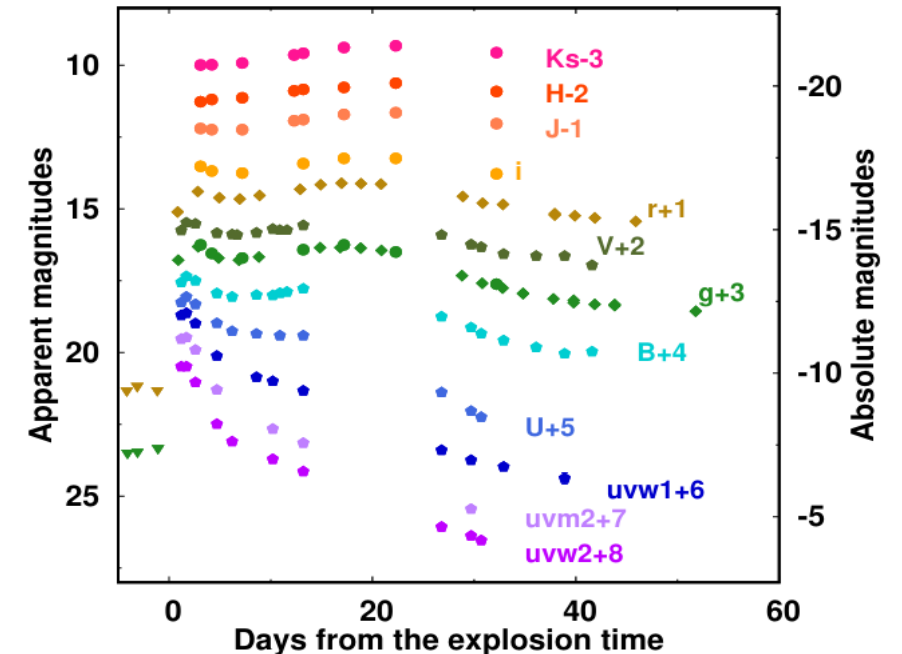
- **instrument development**
- **Supernovae and transients**
 - ⇒ follow-up observation with 1m telescope and 1.4m IRSF (Sutherland, South Africa)
- What is our interest ...
 - **Interacting SNe** (SN IIn/Ibn)
 - **NIR feature in SN IIP**



A growing number of results from kSIRIUS observations !!

- Yamanaka+ 2023, PASJL (SN 2023ixf)
- Yamanaka+ 2025, PASJL (SN 2024iss)
- Yamanaka & Nagayama 2025, ApJL (SN 2023xgo)
- Goto+ 2025, ApJ (SN 2023vbg)

Yamanaka+ 2025 (SN 2024iss)



Outline of this talk

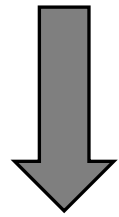
- Iriki Observatory & Our Group
 - introduction our group
- What is a 09ip-like supernova?
- SN 2023vbg
 - Discovery and basic properties
- Light curve evolution
- Blackbody fitting (SED evolution)
- Spectral evolution
- Discussion

What is 09ip-like SNe?

Prototype : SN 2009ip

(e.g, Pastorello+ 2013, Mauerhan+ 2014)

- Pre-SN outburst $\Rightarrow$ **Precursor**
- Fast LC evolution
- Evidence of strong **CSM interaction**



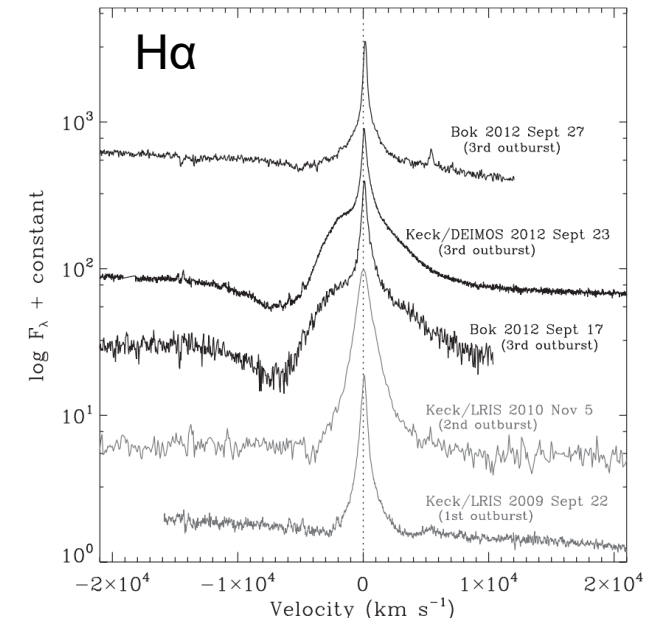
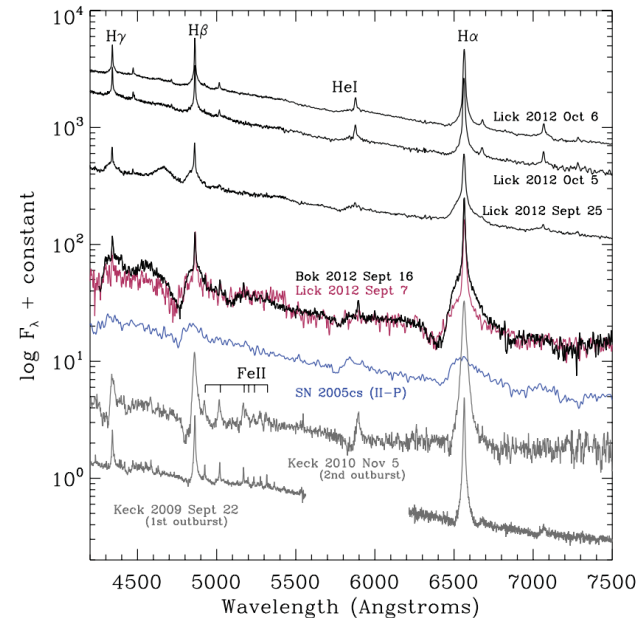
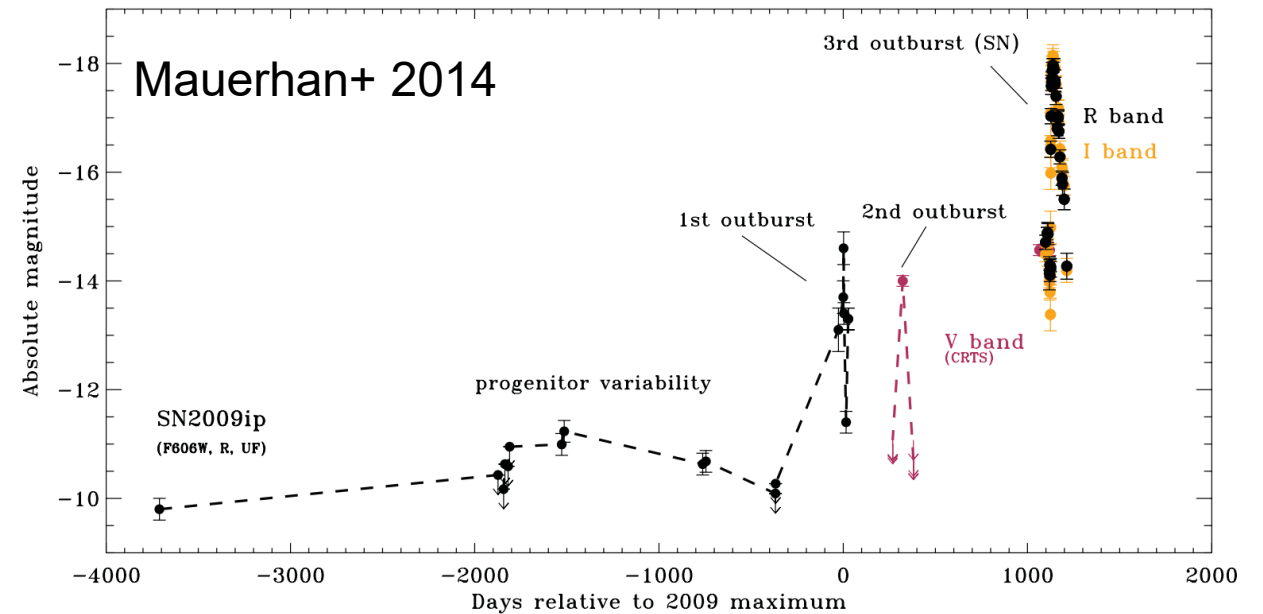
Similar events had
been discovered this 15 years

2009ip-like SNe (09ip-like)

e.g, SN 2010mc, SN 2015bh etc...

- Precursor shortly before explosion
- LC and spectra are very similar to SN 2009ip

Very uniform characteristics!!



What is 09ip-like SNe?

Why 2009ip-like SNe are interesting

➤ Precursor emission

→ A direct probe into final-stage massive stars

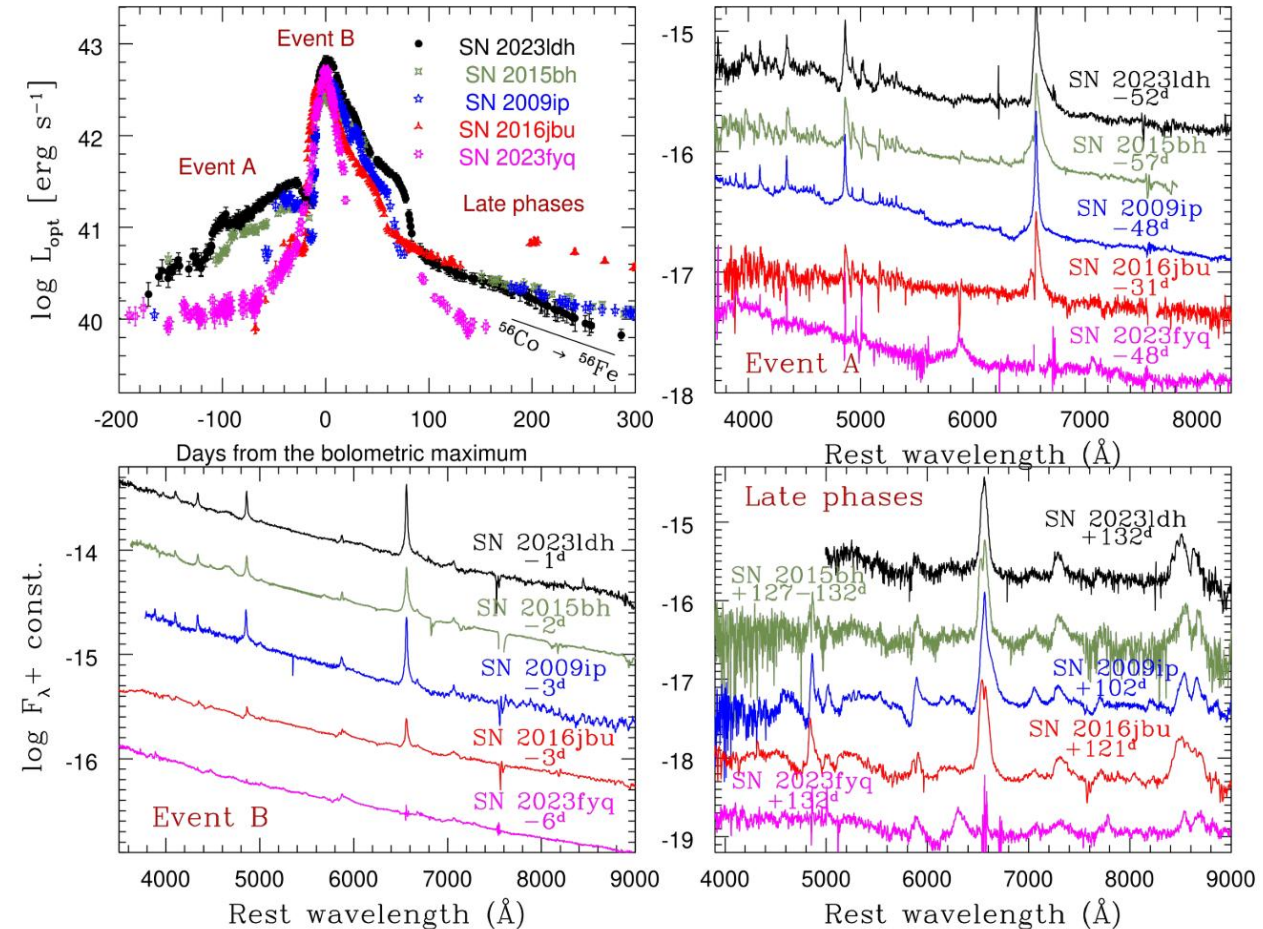
➤ CSM properties

→ Strongly asymmetric and extended CSM
→ Dominates spectra and light curves

➤ Open questions

→ Terminal vs. non-terminal explosion
→ Origin of the apparent uniformity

Pastorello+ 2025 (2023ldh)



- Key objects linking stellar evolution and supernovae
- We need detailed observations and physical modeling

SN 2023vbg

Target : SN 2023vbg

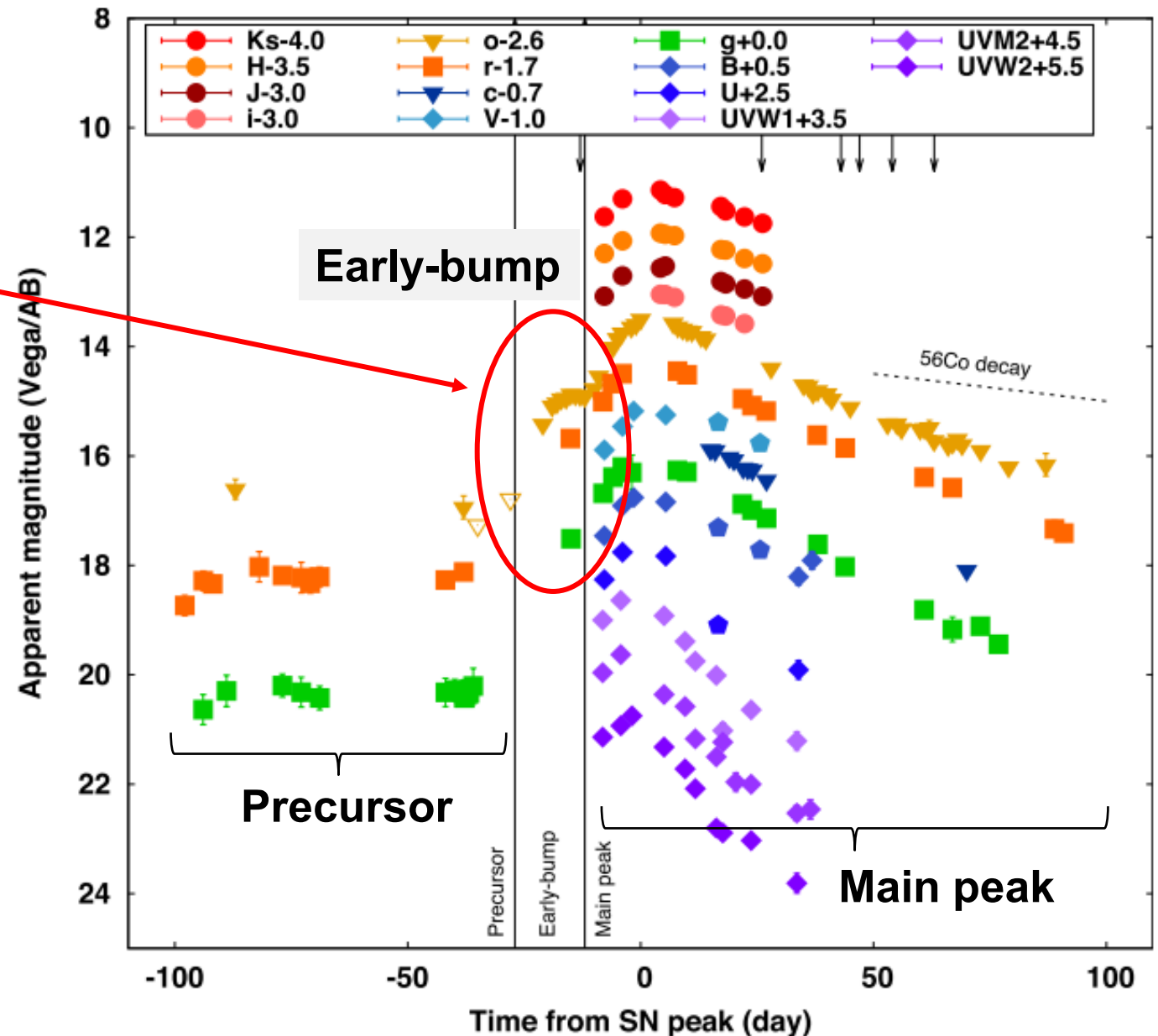
- Discovered by ZTF on 2023 Oct 15 at $r\text{-ZTF} = 20.4$ mag
- **Remained at ~ 20 mag for ~ 80 days $\Rightarrow$ Precursor**
- Fast rising detected by ATLAS on 2023 Dec 30 (orange-ATLAS = 17.2 mag)
- Spectroscopic classification as a **Type II_n SN** at $z = 0.0173$
(Based on observations with Lick 3m / KAST on 2024 Jan 8)
- Follow-up observations initiated using the 1 m telescope (Kagoshima), 2 m HCT (IIA), 3.6 m DOT (ARIES), and Seimei telescope (Kyoto)
 $\Rightarrow$ **5 spectra and JHKs photometry over 9 nights in 60 days**
- Public data from ZTF, ATLAS, and Swift/UVOT were also included

Light curve evolution

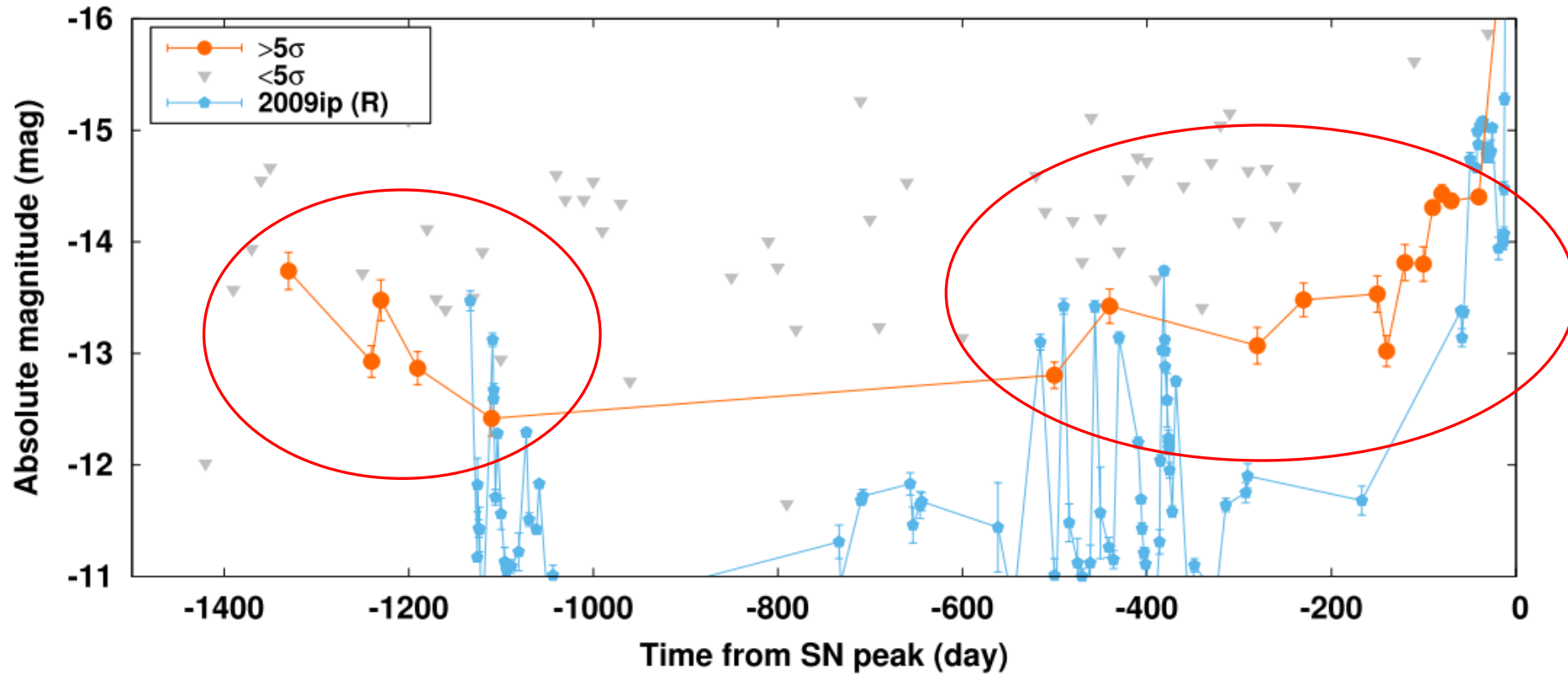
- Precursor lasting ~80 days
- Fast rise + early bump structure
- Main peak
 - Peak magnitude ≈ 16 mag
 - linear decline after peak

Phases defined by the LC morphology
($t=0$ is o-band peak)

- > **Precursor:** $t < -27$
- > **Early-bump:** $-27 < t < -12$
- > **Main peak:** $t > -12$



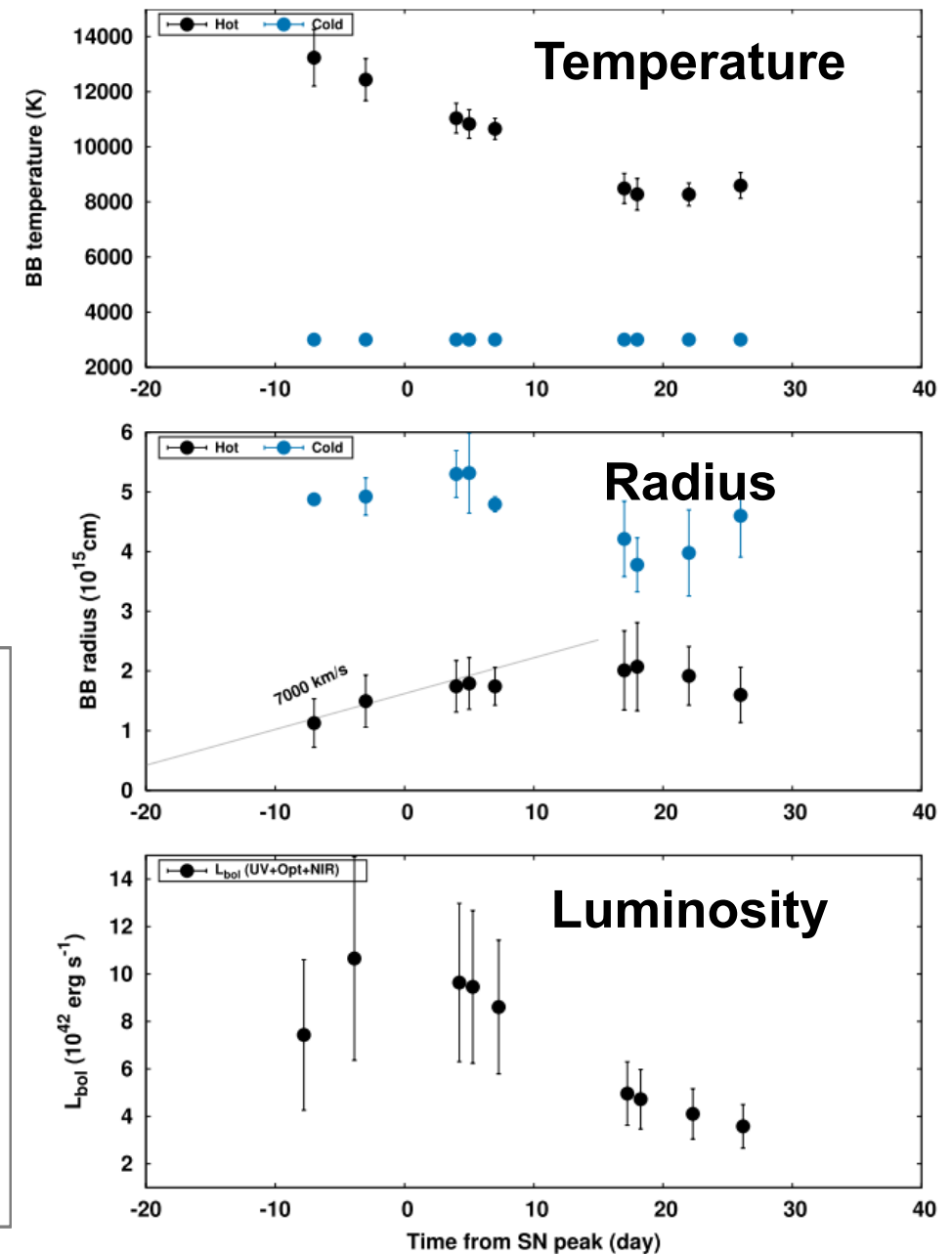
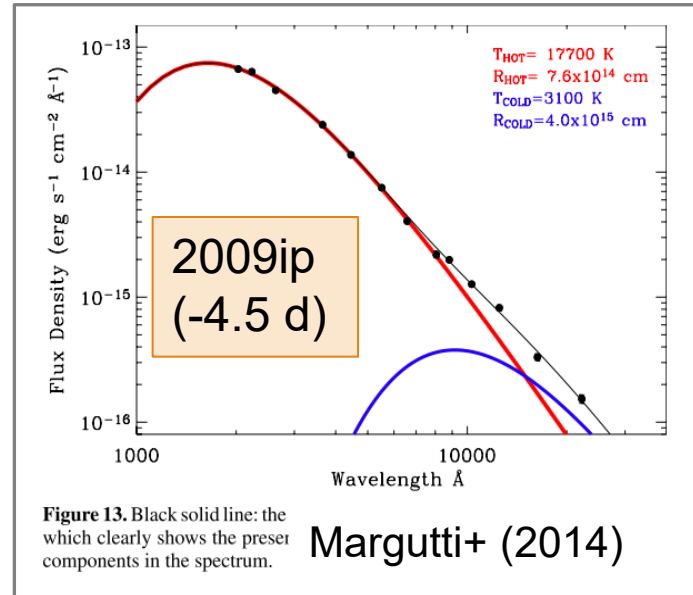
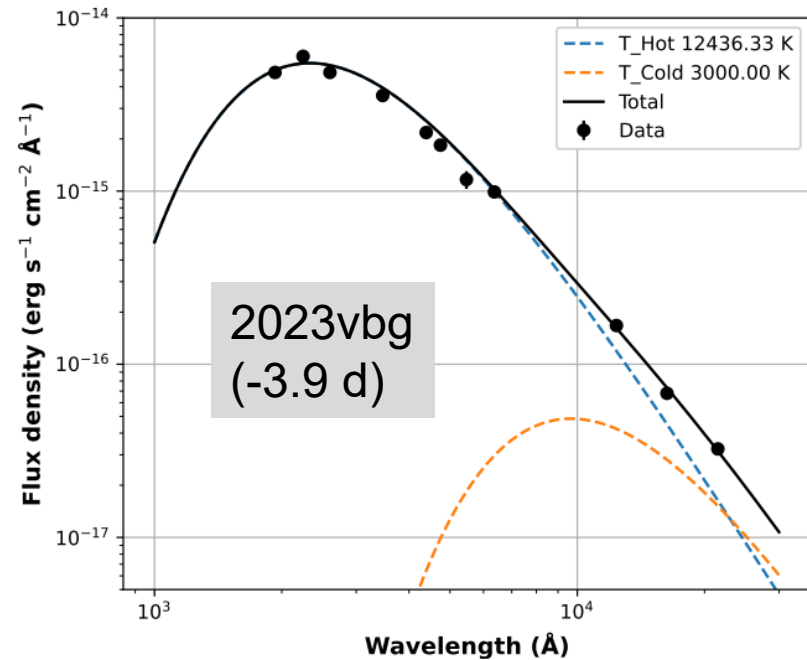
Light curve evolution (precursor phase)



- Archival search (ZTF forced photometry) back to -2000 days
- Possible precursor detections at -1400 to -1000 d and from -400 d
- Brightness comparable to SN 2009ip precursors

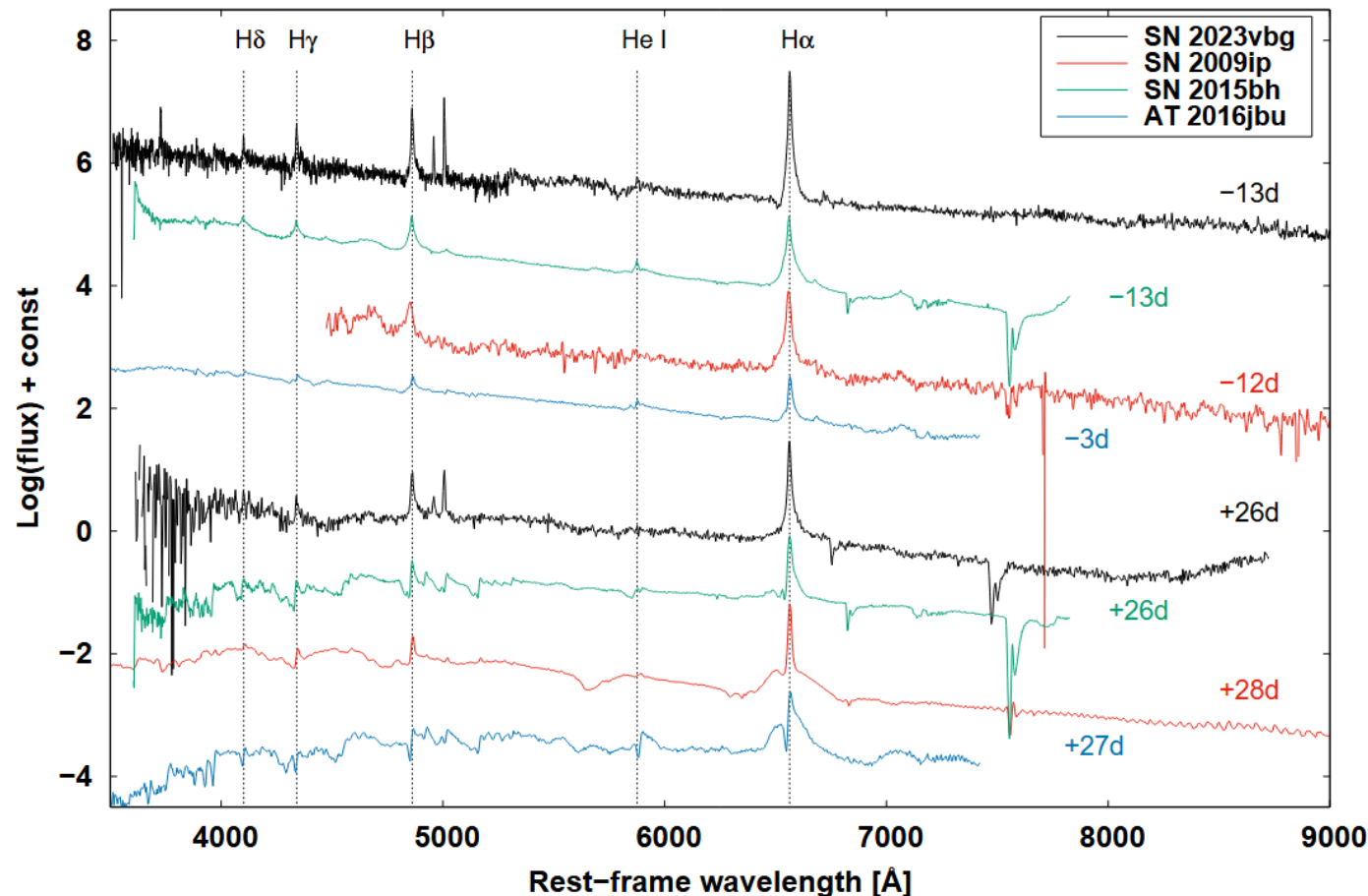
Blackbody fitting (SED evolution)

- BB fitting for SED (UV to NIR)
- Additional cool (3000 K) component is needed?
- Similar excess seen in SN 2009ip (Margutti+2014)
- The origin is still unclear



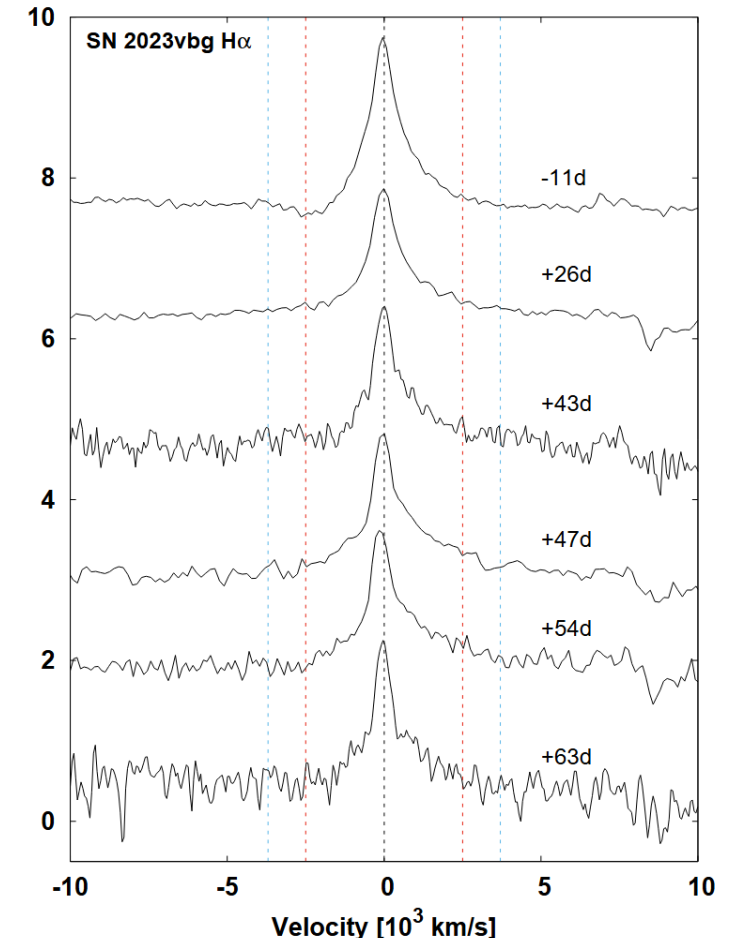
Spectral evolution

- The spectra are dominated by Balmer emission, with only shallow He I $\lambda 5876$ absorption in first spectrum
- The overall spectral evolution shows little change

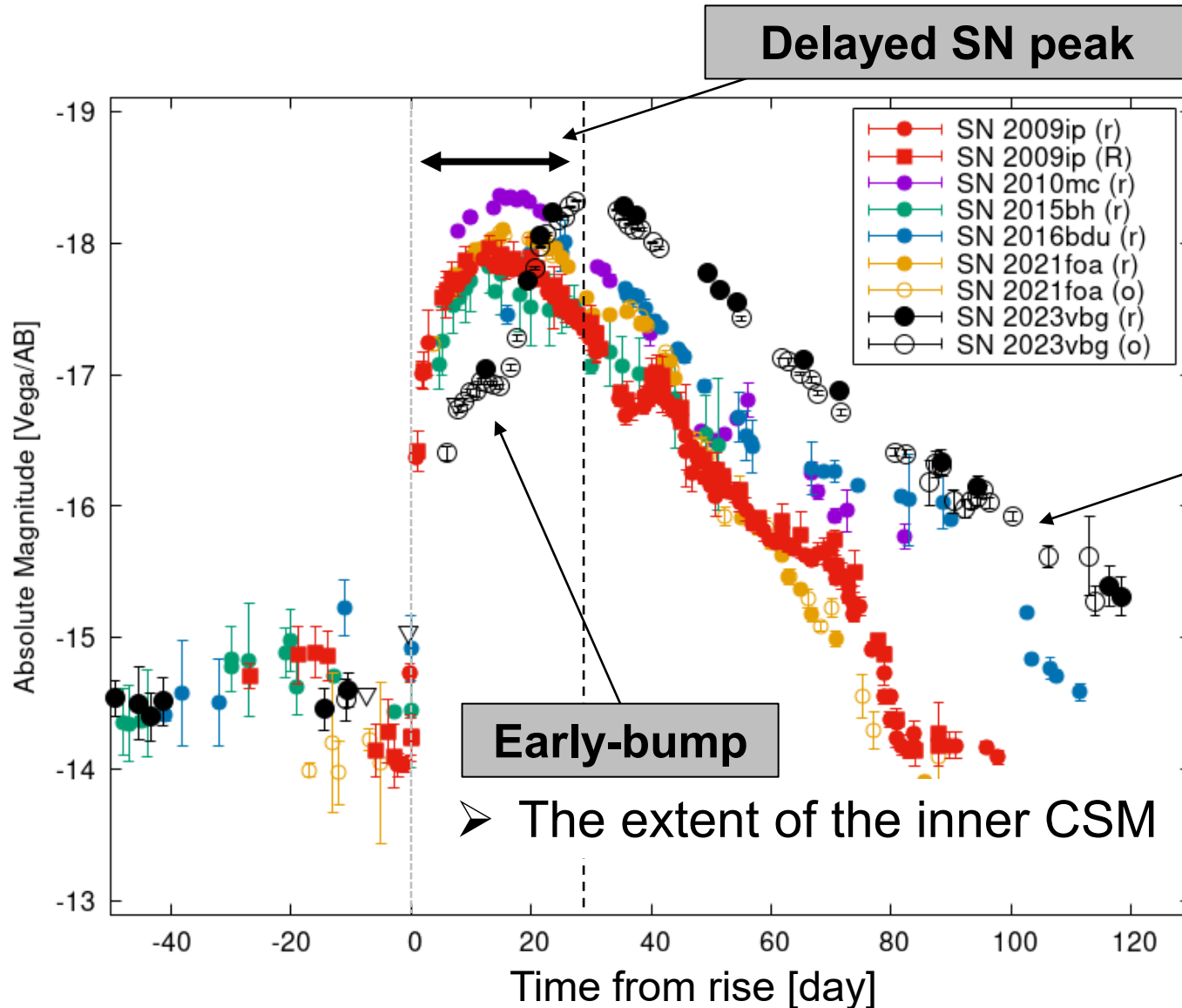


H α : **Narrow** + **Intermediate**
 ~ 500 km/s ~ 3000 km/s

Unlike other 09ip-like SNe,
no broad component is observed



Comparison with 09ip-like SNe



- CSM is dense and extended
⇒ Diffusion timescale is long

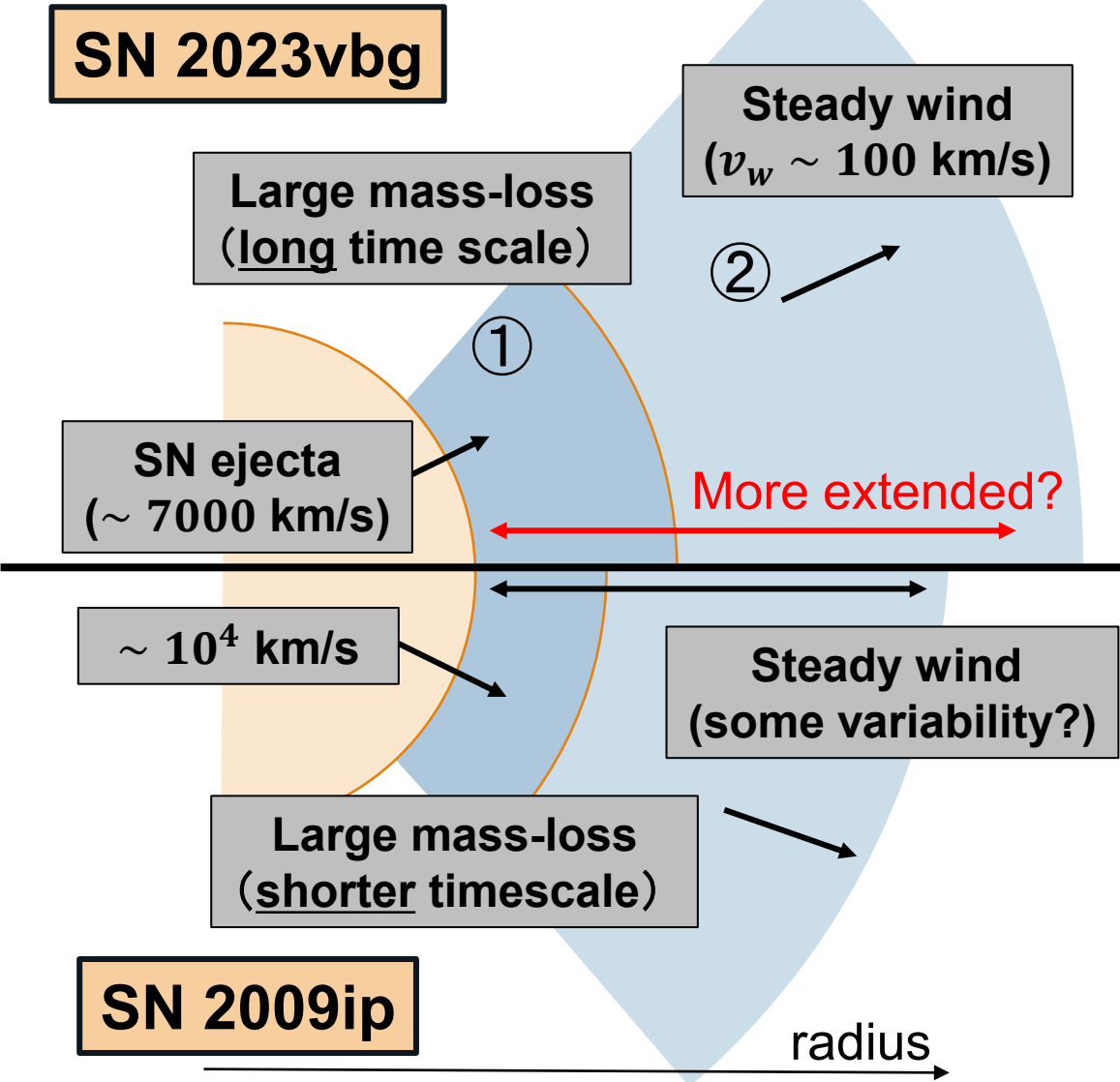
Smooth decline

- No bumpy structure
⇒ uniform density profile
- More extended CSM?

Early-bump

- The extent of the inner CSM

Nature of SN 2023vbg



Two-components CSM

①	②
Inner CSM	Outer wind
Extended, Dense $\Rightarrow$ Early-bump $\bigcirc$	<u>Delayed</u> SN peak
Dense $\Rightarrow$ Early-bump $\times$	Typical SN peak

It is necessary to investigate the origin of the inner, dense CSM.

Summary

- SN 2023vbg is a **new example of a 2009ip-like SN** showing 80 days precursor, discovered by ZTF
- The light curve exhibits a **distinctive bumpy structure** during the rising
- Precursor emission comparable to SN 2009ip is detected as early as ~1400 days before explosion
- Blackbody fitting suggests an **additional cool component** (~3000 K), but its physical origin remains unclear
- The spectra show **only Balmer-series** features, with **no broad component** in H α
- The differences in LC and spectra likely reflect **variations in the pre-explosion CSM structure**, suggesting that SN 2023vbg was surrounded by a more extended CSM

SN 2023vbg is an important sample for exploring the diversity of 09ip-like SNe !!!